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TQC

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Mingsheng Ying "Foundations of Quantum Programming"

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Quantum imperative Programming language

$S :: \text{skip}$

$\text{expression} \rightarrow u := t$ *does nothing*
assignment

$S_1; S_2$ *Bool expr.* *sequencing*

if b then S_1 else S_2 fi *Conditional*

while b do S_1 od *loop construct*

Generalisation of conditional

$\text{if } G_1 \rightarrow S_1$

$\square G_2 \rightarrow S_2$

...

$\square G_m \rightarrow S_m$

fi

$\text{if}(\square_i G_i \rightarrow S_i) \text{ fi}$

*only the first
valid option
will be taken*

*(order is
important!)*

quantum variables ranged over by $q, q', q_1, \dots$

$q \in \text{qVar}$ H_q Hilbert space of dimension 2

$q \in H_q$ (H_∞ infinite dimensionally Hilbert space)

$\bar{q}$ denotes a quantum register $q_1 \dots q_n$
(a finite sequence of distinct variables)

$$\bar{q} \in H_{\bar{q}} = \bigotimes_{i=1}^n H_{q_i}$$

$|\psi\rangle_{q_i}$ denotes that $|\psi\rangle$ is a state of quantum variable q_i :
i.e. $|\psi\rangle \in H_{q_i}$

$|\psi\rangle_{q_i} \langle \varphi|$ outer product of states $|\psi\rangle$ and $\langle \varphi|$ of q_i

$|\psi_1\rangle_{q_1} \dots |\psi_n\rangle_{q_n}$ is a state of the quantum register $\bar{q}$
such that each q_i is in state $|\psi_i\rangle$

For every $i \in [1, n]$

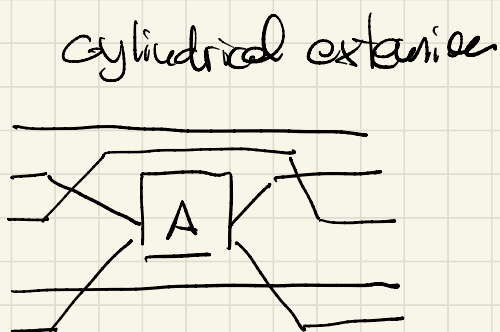
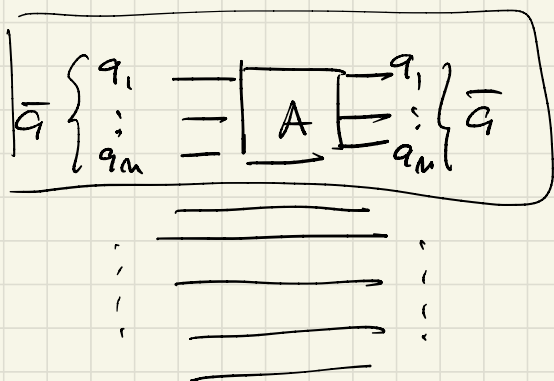
$H_{\text{all}} = \bigotimes_{q \in \text{qVar}} H_q$ Hilbert space of all quantum variables

$\bar{q} = q_1 \dots q_m$ a quater register

A any operator in $H_{\bar{q}}$ has a cylindrical extension

$A \otimes I$ in H_{all}

where I is the identity op in $\bigotimes_{q \in \bar{q} \setminus \bar{q}} H_q$



Quantum programs

$S := \text{skip}$

Note

$q := |0\rangle$

sets q to the initial state $|0\rangle$

$\bar{q} := U[\bar{q}]$

U unitary transformation that we apply to register $\bar{q}$

$S_1; S_2$

$\text{if } (\bigvee_m M[\bar{q}] = m \rightarrow S_m) \text{ fi}$

outcome

quantum case statement

while $M[\bar{q}] = 1$ do S od

quantum loop
yes=1 no=0

Note

For any pure state $|\psi\rangle \in \mathcal{H}_q$ there is a unitary map U on $\mathcal{H}_q$ such that

$|\psi\rangle = U|0\rangle$ so the system can be programmed in a state $|\psi\rangle$ by

$q := |0\rangle;$

$q := U[q]$

About case statement

to see if G_i is satisfied the only way we have is by performing a measurement on it

$$M = \{ M_m \}_{\substack{m \\ \uparrow \text{outcome}}} = \{ M_{m_1}, \dots, M_{m_n} \}$$

$$\sum_m M_m^* M_m = I_d$$

$$\underline{i \notin \left(\bigcap_m M[\bar{a}] = m \rightarrow S_m \right) \underline{p_i}}$$

outcome

probability of the outcome m

$$p(m) = |M_m |\psi\rangle|^2 = \langle \psi | M_m^* M_m | \psi \rangle$$

$$|\psi_m\rangle = \frac{M_m |\psi\rangle}{\sqrt{p(m)}}$$

Difference w.r.t. classical con!

the state of the program is (can be) changed
after performing the measurement by
the collapsing of some qbits!

The same applies to the quantum loop statement

Control flow is classical! "quantum data,
classical flow" model

The outcome of a measurement is always
a classical information

Quantum configurations

$\langle S, \rho \rangle$

Set of all partial density ops in H_{all}
① S is a quantum program or the empty program ϵ

② $\rho \in D(H_{\text{all}})$ is a partial density operator
 $\text{tr}(\rho) \leq 1$ on H_{all} (denotes the global state
of quantum variables)
(loops may not terminate)

SMALL STEP SEMANTICS

$\langle S, p \rangle \rightarrow \langle S', p' \rangle$ denotes a transition from the state p where we still need to execute S to a next state p' where S' remains to be executed

if $S' = \epsilon$ we say the program terminates in state p'

$$\langle \epsilon, p' \rangle \not\rightarrow$$

SOS inference rules

premises

conclusion

----- axioms
conclusion

(Sk)

$$\langle \text{skip}, p \rangle \rightarrow \langle \varepsilon, p \rangle$$

(IN)

$$\langle q := 10, p \rangle \rightarrow \langle \varepsilon, p^q \rangle$$

$$q_0^n = \underbrace{|0\rangle_q \langle 0|}_{} \otimes \underbrace{|0\rangle_q \langle 0|}_{} + \underbrace{|0\rangle_q \langle 1|}_{} \otimes \underbrace{|1\rangle_q \langle 0|}_{} \quad \text{cylindrical extension}$$

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}_{|0\rangle} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}_{\langle 0|} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad A$$

$$\begin{array}{cc} 1 & 0 \\ 0 & 0 \end{array} \quad \begin{array}{cc} 1 & 0 \\ 0 & 0 \end{array} \quad \begin{array}{cc} 1 & 0 \\ 0 & 0 \end{array} \quad \ddots$$

cylindrical extension of A

(UT)

$$\frac{\langle \bar{a} := U[\bar{a}], P \rangle}{\langle \varepsilon, U P U^* \rangle}$$

cylindrical ext.
of U

(SC)

$$\frac{\langle S_1, P \rangle \rightarrow \langle S_1', P' \rangle}{\langle S_1; S_2, P \rangle \rightarrow \langle S_1'; S_2, P' \rangle}$$

(EM)

$$\frac{\langle S_2, P \rangle \rightarrow \langle S_2', P' \rangle}{\langle \varepsilon; S_2, P \rangle \rightarrow \langle S_2', P' \rangle}$$

Alternatively

$$\varepsilon; S_2 = S_2$$

$$P_m \neq 0$$

(IF)

$$\langle \text{if } (\bigsqcup_m M[\bar{a}] = m \rightarrow S_m) \& i, P \rangle \rightarrow \langle S_m, \underline{M_m P M_m^*} \rangle$$

cylindrical
ext.
↓

is a nondeterministic rule!

P_m = $\text{tr}(M_m P M_m^*)$ is the probability of
nond choice

(L0)

$$P_0 \neq 0$$

$$\langle \text{while } M[\bar{q}] = 1 \text{ do } S \text{ od}, P \rangle \rightarrow \langle E, M_0 P M_0^* \rangle$$

$$P_0 = \text{tr}(M_0 P M_0^*)$$

↖ ↗
cylindrical ext

(L1)

$$P_i \neq \emptyset$$

$$\langle \text{while } M[\bar{q}] = 1 \text{ do } S \text{ od}, P \rangle$$

$$\rightarrow \langle S; \text{while } M[\bar{q}] = 1 \text{ do } S \text{ od}, M_i P M_i^* \rangle$$

$$P_i = \text{tr}(M_i P M_i^*)$$

Alternatively:

$$P_m = \text{tr}(M_m P M_m^*) \neq 0 \quad P_m = \frac{M_m P M_m^*}{P_m}$$

$$\langle \text{if } \bigwedge_m M[\bar{c}] = m \rightarrow S_m \text{ fi}, P \rangle \xrightarrow{P_m} \langle S_m, P_m \rangle$$

(and similarly for loops)

$$\sum P_i = 1$$



Denotational Semantics

- we write $\langle S, P \rangle \xrightarrow{m} \langle S', P' \rangle$
iff we reach $\langle S', P' \rangle$ from $\langle S, P \rangle$ in exactly m transitions

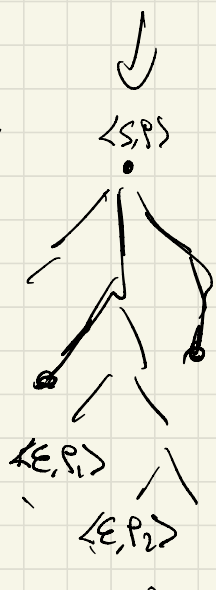
- we write $\langle S, P \rangle \xrightarrow{*} \langle S', P' \rangle$
iff $\exists m \in \mathbb{N} \quad \langle S, P \rangle \xrightarrow{m} \langle S', P' \rangle$

math world

$$\llbracket S \rrbracket : D(\text{Hull}) \rightarrow D(\text{Hull})$$

↑
syntax

$$\llbracket S \rrbracket P = \sum \left\{ \underline{P'} \mid \langle S, P \rangle \xrightarrow{*} \langle E, P' \rangle \right\}$$



Lemma $\llbracket S \rrbracket$ is a linear function

proof by structural induction on S

Th

$$\llbracket \text{while } \dots \rrbracket \rho = M_0 \rho M_0^* + \llbracket \text{while} \rrbracket (\llbracket S \rrbracket (M_1 \rho M_1^*))$$

where $\text{while} = (\text{while } \neg \llbracket \varphi \rrbracket = 1 \text{ do } S \text{ od})$

Proposition

$$\text{tr}(\llbracket S \rrbracket \rho) \leq \text{tr}(\rho)$$

intuitively, $\text{tr}(\llbracket S \rrbracket \rho)$ is the probability that
the program S terminates
on ρ

$\text{tr}(\rho) - \text{tr}(\llbracket S \rrbracket \rho)$ is the probability of
divergence of
 S on ρ

Grover Quantum search

$N = 2^m$ elements

exactly L solutions

$$1 \leq L \leq \frac{N}{2}$$

- 1 $q_0 := |0\rangle; \dots q_{m-1} := |0\rangle;$
- 2 $q := |0\rangle;$
- 3 $r := |0\rangle;$ count the number of iterations (classical var)
- 4 $q := X[q];$
- 5 $q_0 := H[q_0]; \dots q_{m-1} := H[q_{m-1}];$
- 6 $q := H[q];$
- 7 while $M[r] = 1$ do $\triangleright$ ok
- 8 if $(\bigcap_z M'[q_0 \dots q_{m-1}] = x \rightarrow \text{skip})$ fi

Measurement M

$$M_0 = \sum_{\ell \geq k} |\ell\rangle_{\pi} \langle \ell| \quad M_1 = \sum_{\ell < k} |\ell\rangle_{\pi} \langle \ell|$$

$$M' = \{M'_x : x \in \{0, 1\}^n\} \quad M'_x = |x\rangle \langle x|$$

loop body D:

$$q_0 \dots q_{m-1} := \overset{\text{oracle}}{\mathcal{O}}[q_0 \dots q_{m-1}];$$

$$q_0 := H[q_0]; \dots; q_{m-1} := H[q_{m-1}];$$

$$q_0 \dots q_{m-1} := \underline{P_h}[q_0 \dots q_{m-1}] \quad \leftarrow \begin{array}{l} \text{conditional} \\ \text{phase shift} \end{array}$$

$$q_0 := H[q_0]; \dots; q_{m-1} := H[q_{m-1}];$$

$$|0\rangle \rightarrow |0\rangle$$

$$|x\rangle \rightarrow -|x\rangle \neq 0$$

$$r := r + 1$$